

PREDICTION: LEGITIMATE

RISK SCORE: 58 / 100 (MEDIUM RISK)

Scam Probability: 40.09% | Model Confidence: 59.91%

Banking Action: OTP VERIFICATION

Official Recommendation: OTP Verification Required. Moderate risk parameters detected. Challenge customer with SMS OTP or biometric 2FA authentication.

1. Transaction Details & Actual Parameter Values

Transaction ID:	TXN-VERIFY-SCAM-001	Transaction Amount:	\$95,000.00
Customer ID:	C999888777	Sender Old Balance:	\$95,000.00
Beneficiary ID:	M111222333	Sender New Balance:	\$0.00
Payment Type:	CASH_OUT	Receiver Old Balance:	\$0.00
Device Type:	Unknown Proxy	Receiver New Balance:	\$95,000.00
Location Region:	Foreign Proxy	6h Transaction Velocity:	12 txns
New Beneficiary:	Yes (New)	Prior Fraud History:	Yes (High Risk)

2. Composite Risk Score Breakdown & Latency Timing

Risk Dimension	Sub-Score	Status & Impact
Model Fraud Probability	40.09%	Weight: 70% (Score: 40.1)
Amount Risk	35.0 pts	Transaction size evaluation
Velocity Risk	35.0 pts	6-hour transaction frequency
Device Risk	25.0 pts	Device profile verification
Location Risk	25.0 pts	Geographic regional risk
Beneficiary Risk	15.0 pts	New beneficiary check
Historical Behaviour	30.0 pts	Prior fraud flag check
Balance Behaviour	35.0 pts	Account wipeout detection
FINAL COMPOSITE RISK SCORE	58 / 100	MEDIUM RISK CATEGORY

Feature Eng: 28.05 ms	Prediction: 1.17 ms	SHAP Engine: 1.71 ms	LIME Engine: 119.30 ms	Rendering: 1110.35 ms	TOTAL: 1286.09 ms
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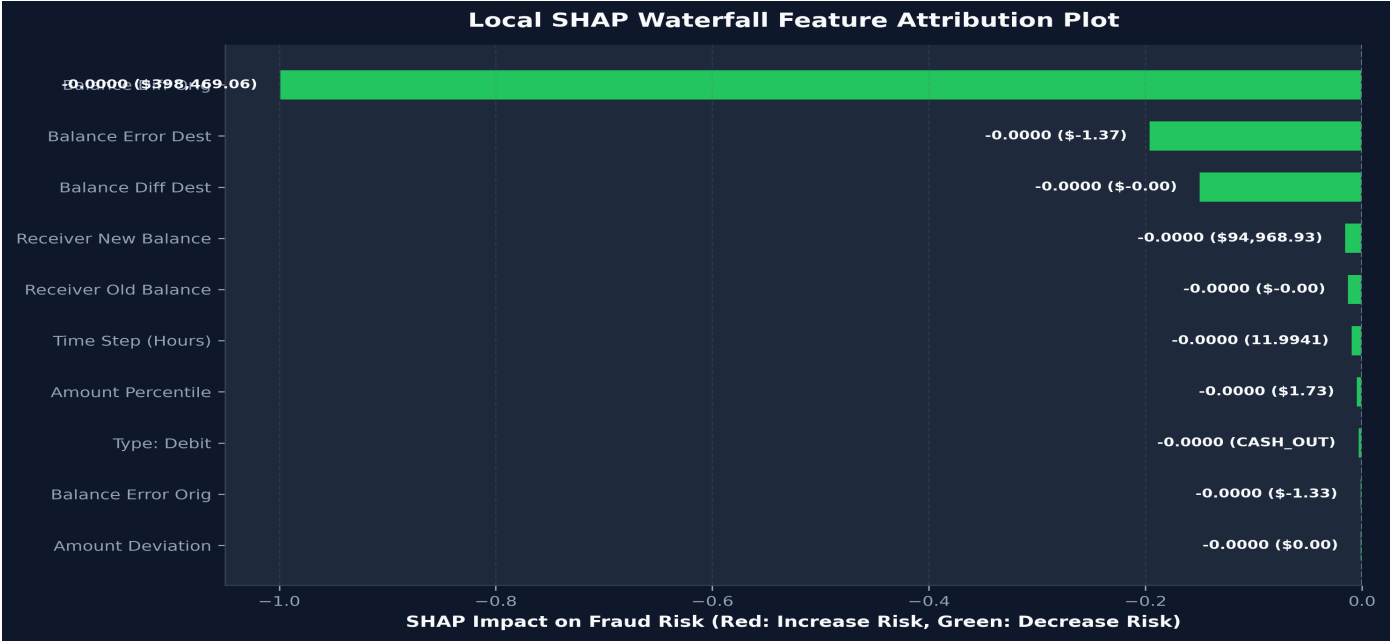
3. Banking-Grade Analyst Narrative Report

Transaction Summary:	Transaction TXN-VERIFY-SCAM-001 involving Customer C999888777 transferring \$95,000.00 via CASH_OUT to Beneficiary M111222333 on Unknown Proxy from Foreign Proxy.
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Behaviour Analysis:	Complete account balance drain: Sender balance wiped out from \$95,000.00 to \$0.00. High velocity anomaly: 12 transactions executed within the last 6 hours. Device anomaly: Payment initiated from unverified device profile ('Unknown Proxy'). Geographic anomaly: Payment originated from high-risk or foreign region ('Foreign Proxy').
High Risk Indicators:	Elevated Transaction Amount: \$95,000.00 exceeds single transfer threshold.; Sender Account Wipeout: Entire origin balance drained to zero.; Rapid Transaction Spike: 12 transfers within a 6-hour window.; Unverified Device Profile: Initiated via Unknown Proxy.; Suspicious Geographic Location: Access from Foreign Proxy.; Unestablished Beneficiary: No prior transfer history with recipient account.; Beneficiary Risk Flag: Recipient account associated with prior fraud incidents.; Off-Hours Activity: Transaction executed during late-night window.
Safe Indicators:	None identified.
Model Explanation:	GuidedGuard Gradient Boosting model evaluated this payment pattern with a scam probability of 40.09% and classified it as 'Legitimate'. The composite banking risk score is categorized under MEDIUM RISK.
SHAP Summary:	Feature contributions align with baseline legitimate profile.
Final Recommendation:	OTP Verification Required: Moderate risk parameters detected. Challenge customer with SMS OTP or biometric 2FA authentication.

4. Top 10 SHAP Feature Contribution Attributions

Feature Name	Actual Value	SHAP Value	Contrib %	Direction	Impact Level
Balance Diff Orig	\$398,469.06	-0.0000	45.47%	Decreases Risk	High
Balance Error Dest	\$-1.37	-0.0000	8.95%	Decreases Risk	Medium
Balance Diff Dest	\$-0.00	-0.0000	6.86%	Decreases Risk	Medium
Receiver New Balance	\$94,968.93	-0.0000	0.75%	Decreases Risk	Low
Receiver Old Balance	\$-0.00	-0.0000	0.62%	Decreases Risk	Low
Time Step (Hours)	11.9941	-0.0000	0.48%	Decreases Risk	Low
Amount Percentile	\$1.73	-0.0000	0.25%	Decreases Risk	Low
Type: Debit	CASH_OUT	-0.0000	0.19%	Decreases Risk	Low
Balance Error Orig	\$-1.33	-0.0000	0.11%	Decreases Risk	Low
Amount Deviation	\$0.00	-0.0000	0.10%	Decreases Risk	Low



5. Local LIME Decision Rules & Explanations

Decision Rule Condition	Rule Weight	Effect Direction	Domain Explanation
Sender Old Balance = \$95,167.89	+11420.0866	Increases Risk	Feature condition 'Sender Old Balance = \$95,167.89' contributed positively to scam classification.
Receiver New Balance = \$94,968.93	+11396.2117	Increases Risk	Feature condition 'Receiver New Balance = \$94,968.93' contributed positively to scam classification.
Transaction Amount = \$94,812.48	+11377.4376	Increases Risk	Transaction amount is significantly elevated relative to normal threshold.
Time Step (Hours) = 11.9941	+1.3793	Increases Risk	Feature condition 'Time Step (Hours) = 11.9941' contributed positively to scam classification.
Type: Cash Out = CASH_OUT	+0.0604	Increases Risk	Transfer/Cash Out payment type has higher statistical fraud incidence.
Type: Payment = CASH_OUT	-0.0602	Decreases Risk	Payment type associated with routine account transfers.
Type: Transfer = CASH_OUT	-0.0602	Decreases Risk	Payment type associated with routine account transfers.
Receiver Old Balance = \$-0.00	-0.0600	Decreases Risk	Feature condition 'Receiver Old Balance = \$-0.00' supported legitimate transaction verification.
Type: Debit = CASH_OUT	-0.0595	Decreases Risk	Payment type associated with routine account transfers.
Sender New Balance = \$0.00	-0.0594	Decreases Risk	Feature condition 'Sender New Balance = \$0.00' supported legitimate transaction verification.

